

Second Terminal Examination - 2079

Class : XI

Subject: Biology (Code: 2011)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

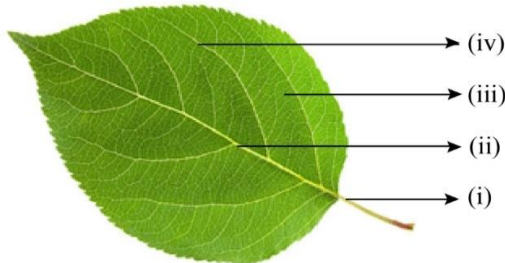
PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet.

[6 × 1 = 6]

1. Endoplasmic reticulum is made up of
 - a. Cisternae, Vesicles & Tubules
 - b. Tubules only
 - c. Tubules & Vesicles
 - d. Cisternae & Vesicles
2. The number ii shown in the diagram below represents.....



- a. Petiole
 - b. Midrib
 - c. Vein
 - d. Lamina
3. The border between forest & grassland is an example of
 - a. Ecotype
 - b. Ecological niche
 - c. Ecotone
 - d. Biome
4. The exchange between the sporophytic and gametophytic generations in the life cycle of bryophyta is called
 - a. Continuity of life
 - b. Life regulation
 - c. Heteromorphic generation
 - d. Alternation of generation
5. The living materials in the cytoplasm that take part in cellular activity are
 - a. Cell organelles
 - b. Ergastic substances
 - c. Desmosomes
 - d. Plasmodesmata.

6. Pyrenoids are
- Protein surrounded by oil droplets
 - Protein surrounded by starch grains
 - Starch surrounded by protein grain
 - Starch surrounded by oil droplets

Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

1. What is the nature of plasma membrane? Write down the functions of plasma membrane [1+3]

OR

Define synecology. Describe the role of light on flora & fauna in any ecosystem. [1+3]

2. Explain the structure and function of golgi complex. [2+2]
3. Name the reserve food material of algae. Distinguish between Chlorophyceae and phaeophyceae. [1+3]
4. On the basis of the diagram given below, answer the following questions. [1×4]

- Name the plant in the figure.
- To what family does the plant belong?
- What type of modified tap root is this?
- What is the main purpose of this type of root?



Group 'C' [BOTANY]

Give long answer to the following questions.

[2 × 8 = 16]

5. Describe the structural and functional aspects of pond ecosystem in detail. [4+4]
6. Describe the family Cruciferae in semi-technical terms with floral formula and floral diagram. State the scientific names of any two economic plants of the family. [4+3+1]

OR

How does *Spirogyra* get its name? Describe the life cycle of *Spirogyra*. [1+7]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet.

[5 × 1 = 5]

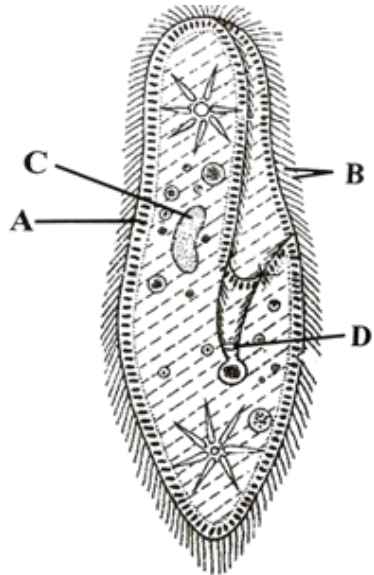
1. Which of the following facts definitely casts doubt on Lamarckian theory:
 - a. Snakes lost their limbs due to burrowing mode of life
 - b. Neck and fore limbs of African giraffe use for generations
 - c. Vestigial organs were functional once but now undergone degeneration due to loss of their function
 - d. Blacksmith's son does not have powerful arm muscles as his father
2. In *Paramecium*, osmoregulation occurs through
 - a. Body surface
 - b. Cilia
 - c. Nucleus
 - d. Contractile vacuole
3. Which of the following substances will not be converted to compost when added in a composting pit?
 - a. Waste paper
 - b. Fruit and vegetable peels
 - c. Human and animal excreta
 - d. Plastic bags
4. Conjugation as a mode of reproduction is present in
 - a. Ciliata
 - b. Flagellata
 - c. Sarcodina
 - d. Sporozoan
5. Which of the following features could not be regarded correct in the line of human evolution:
 - a. Opposable thumb
 - b. Reduced fertility due to increased gestation and single child birth
 - c. Rolled out lips and elevated breasts
 - d. Decrease in intelligence

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 = 16]

1. The following is the diagram of *Paramecium* & answer the following:



- a) Name C & D. [2]
b) What is the function of A and D [2]
2. Define Mutation and explain any two postulates of Neo Darwinism. [1+3]
3. "Aves are glorified reptiles". Explain. [4]
4. Discuss transverse binary fission in *Paramecium* with well labelled diagram. [3+1]

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

5. What is conjugation? Describe conjugation in *Paramecium caudatum* with diagrams. Mention any two significances of conjugation. [1+4+2+1]
6. Name the theory of evolution purposed by Darwin. Enumerate the key concepts in the evolution theory of Darwin. [1+7]

ॐ The End ॐ